Library walking access

Seattle, Washington, USA

42% of residents can reach a library within a 15-minute walk (slope-aware travel time)

415,307 people are beyond that 15-minute standard

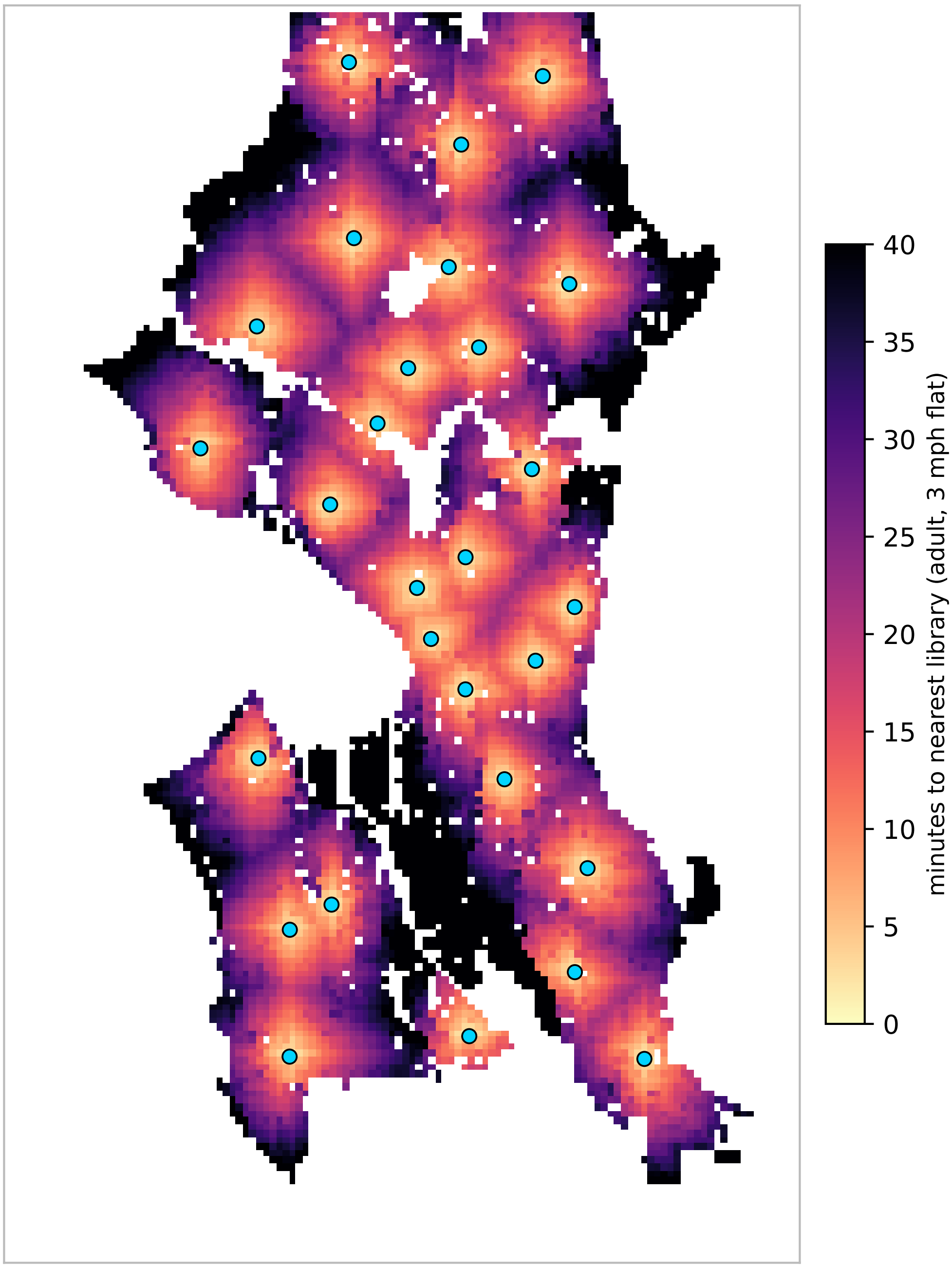
87% of residents with an ambulatory difficulty are beyond a 15-minute walk

43 min is the median walk faced by the worst-served of that group — before any slope penalty at all

Scope 28 library locations · 206 km² analysed · 720,959 residents
Terrain elevation -1–158 m · 13.1% of walk segments steeper than 8.33%
Standard 15-minute neighbourhood goal; 3 mph on the flat = 0.75 mi
Sources OpenStreetMap walk network · US Census ACS 5-year 2023 · USGS 3DEP elevation

Where the walk is long

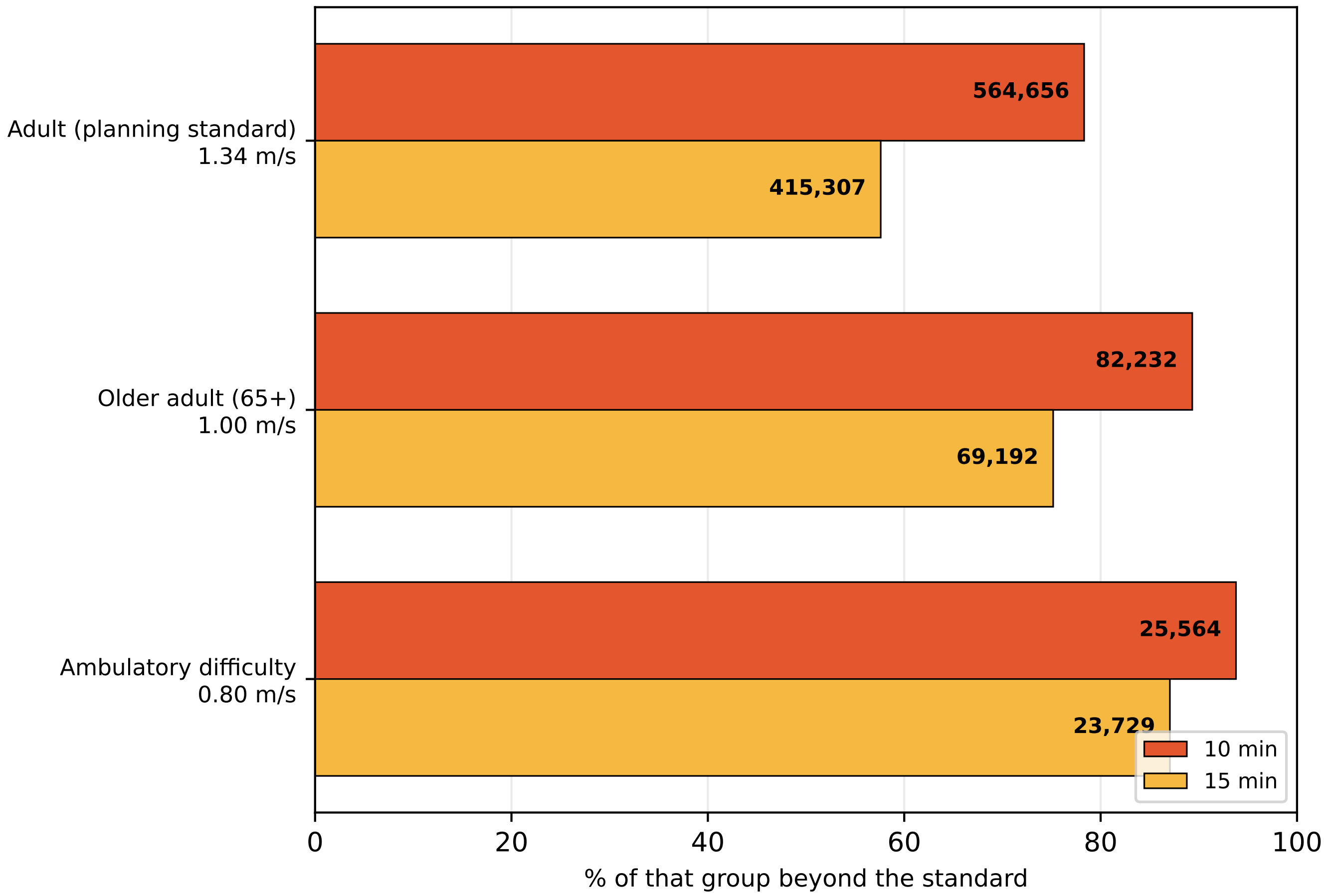
Travel time to the nearest library, on foot, accounting for slope



Median walk is 21 minutes. Blue dots are the 28 branches. Darker areas are further in time, not distance — a steep half-mile costs more than a flat one. Areas outside the analysed land mask (open water, and cells more than 250 m from any mapped pedestrian way) are blank.

A time standard is not one distance

The same policy sentence describes a different city for each kind of walker

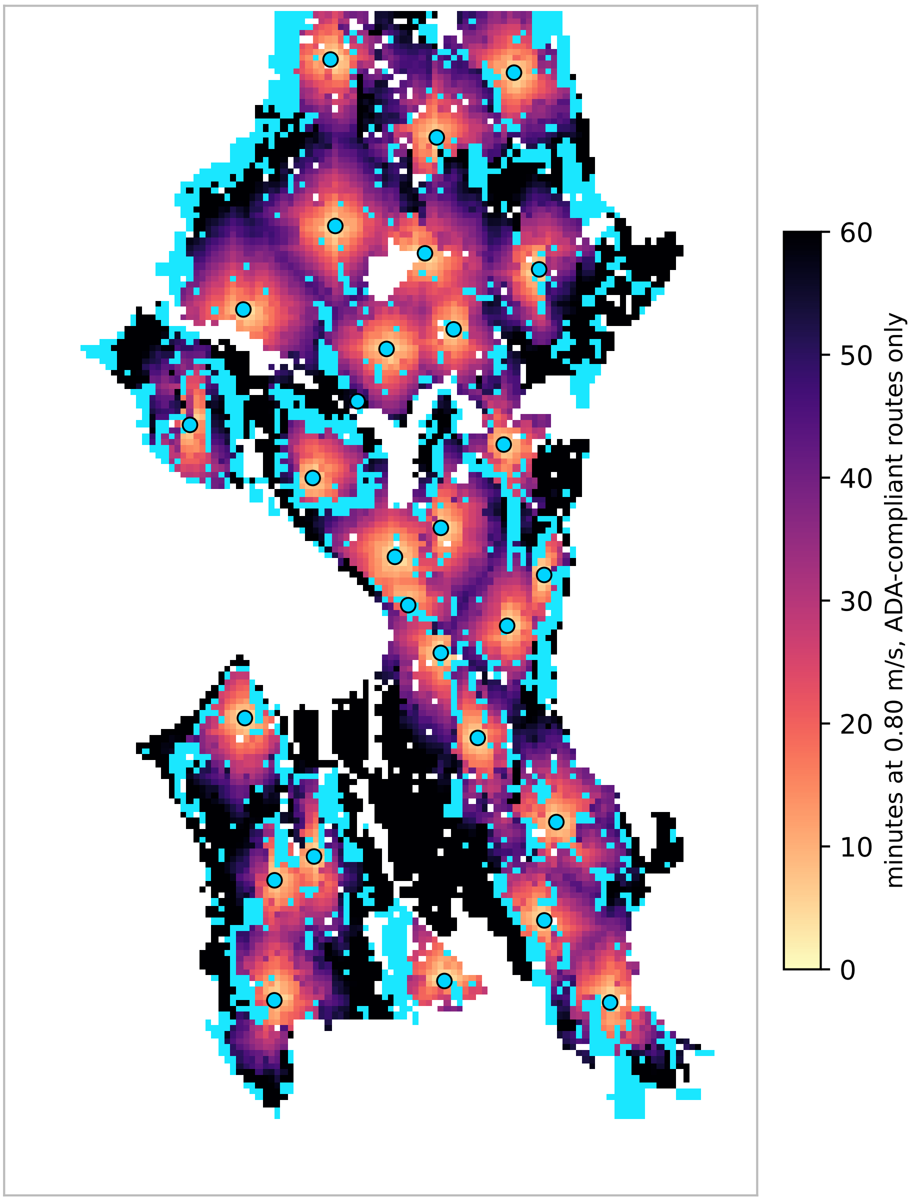


profile	group	10 min	15 min	no route
Adult (planning standard)	720,959	564,656 (78%)	415,307 (58%)	0
Older adult (65+)	92,052	82,232 (89%)	69,192 (75%)	0
Ambulatory difficulty	27,258	25,564 (94%)	23,729 (87%)	4,591

Each row is measured against its own population, not a share of the city. Walking speeds are planning defaults: 3 mph adult, 1.00 m/s for 65+, 0.80 m/s with an ambulatory difficulty.

Slope, and who it excludes

Grade excludes people that distance alone does not, and the effect is concentrated

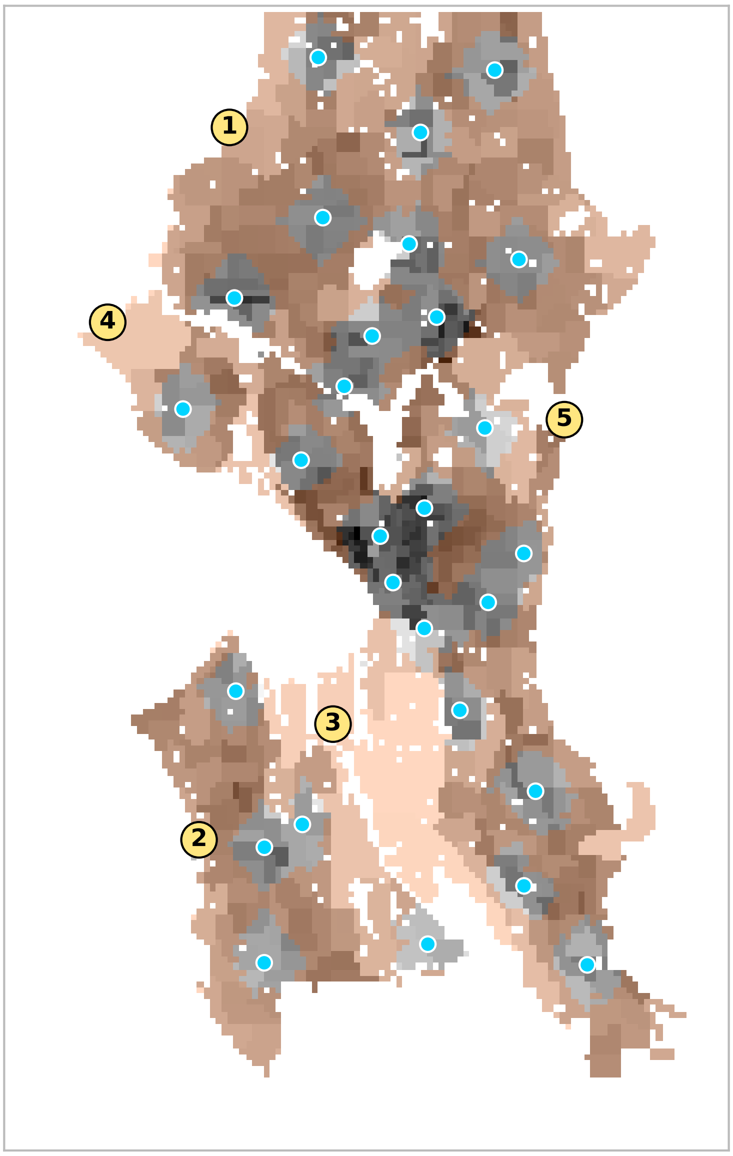


Cyan marks land with no ADA-compliant walking route to any library: 4,591 residents with an ambulatory difficulty, 17% of that group. 13.1% of walk segments exceed the 8.33% limit.

That count is sensitive to the threshold — it ranges 1,008–12,848 between a 1:20 and a 1:8 cutoff — so it should not be read as a precise figure. The finding underneath it is not sensitive. These are remote locations before slope is considered at all: with no slope penalty whatsoever the same cohort still faces a median 43-minute walk, and 18% of them over an hour. Slope is a second effect on top of an already severe access deficit, and it falls almost wholly on wheelchair users.

Underserved pockets

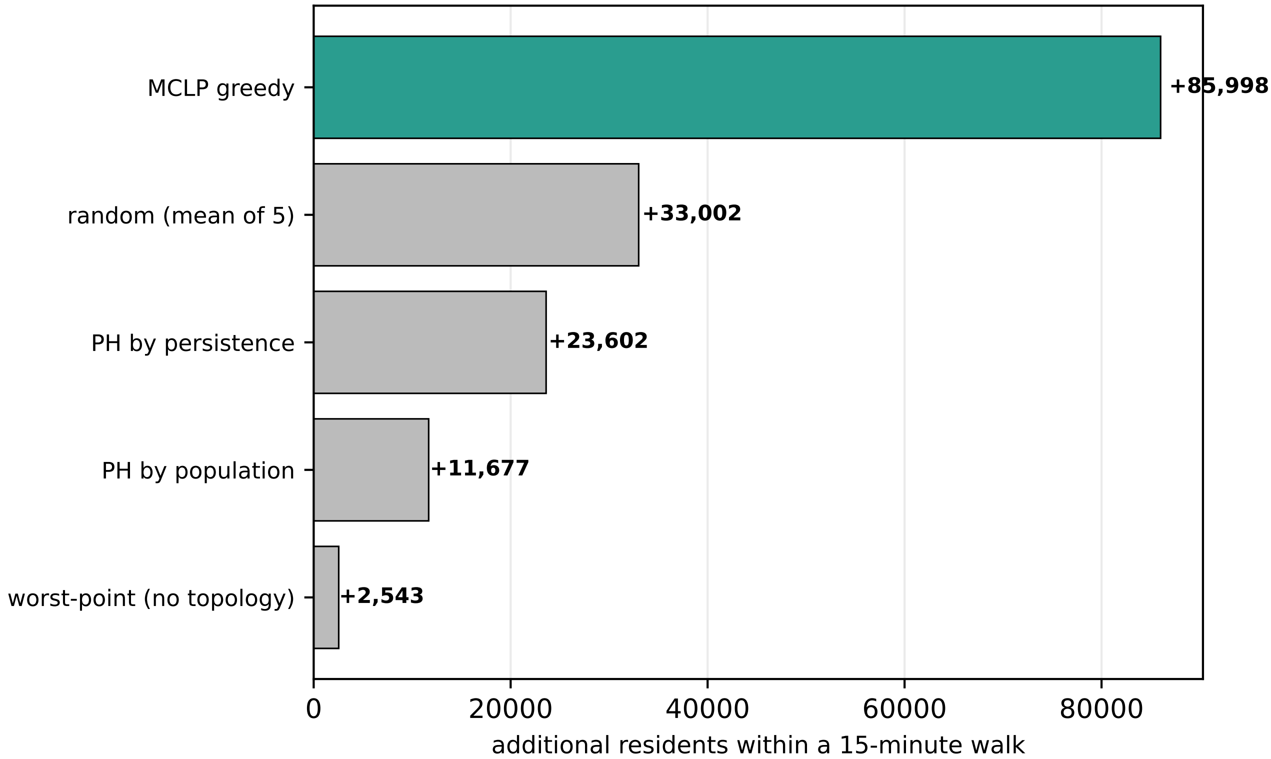
Contiguous areas beyond a 15-minute walk, ranked by residents affected



#	residents	car-free HH	area	worst walk
1	169,032	7,160	55.6 km²	59 min
2	67,998	2,282	27.6 km²	66 min
3	66,867	3,378	35.8 km²	136 min
4	58,103	6,848	15.7 km²	64 min
5	45,321	4,350	9.5 km²	53 min
6	4,253	553	0.9 km²	26 min

Where should the next 8 branches go?

Siting strategies scored on residents brought within 15 minutes



Greedy maximal-covering location (MCLP) adds 85,998 residents with 8 branches, raising coverage from 42% to 54%. It is scored here against topological and trivial alternatives; see the repository for the full comparison. Candidate sites are inhabited cells on a 450 m grid.

Method and caveats

Walk network from OpenStreetMap, retaining all connected components. Travel time uses Tobler's hiking function renormalised to each profile's flat speed; the mobility profile additionally treats segments steeper than 8.33% as impassable. Population is ACS 2023 block-group data rasterised to a 150 m grid; poverty, vehicle access and ambulatory difficulty are tract-level and therefore coarser.

Known limits: sidewalk quality, curb ramps, crossing delay and transit are not modelled, so the mobility figures remain optimistic even with slope. Walking speeds are literature defaults rather than locally observed. Travel time is one-way; a downhill outbound trip is uphill on return. Facility sets are branch locations only and take no account of opening hours or capacity.